

PARTS LIST

Semiconductors:

- IC1 - ATmega8 microcontroller
- IC2 - LM35 temperature sensor
- D1 - 1N4001 rectifier diode
- T1, T2 - BC557 pnp transistor
- T3 - BC547 npn transistor
- DIS1, DIS2 - LTS542 common-anode 7-segment display

Resistors (all 1/4-watt, 45% carbon):

- R1, R2, R12 - 1-kilo-ohm
- R3 - 10-kilo-ohm
- R4-R11 - 330-ohm

Capacitors:

- C1-C3 - 0.1µF ceramic disk
- C4 - 10µF, 16V electrolytic

Miscellaneous:

- CON1 - 3-pin terminal block connector
- CON2 - 6-pin connector
- CON3, CON4 - 2-pin terminal block connector
- RL1 - 5V, 1 C/O relay
- S1 - Push-to-on tactile switch
- Desktop SMPS
- Thermoelectric module
- TEC1-12706
- Desktop CPU fan (12V DC)
- Thermocol box
- Aquatic air-pump

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

IC1. Once the PCB is assembled, power it on with the SMPS as shown in Fig. 2. The DC fan should immediately start rotating. Check the voltage levels at various test points as shown in test point table to ensure the circuit is working as required.

Mechanical construction. The mechanical construction is the most critical part of this project. The overall body of the refrigerator is a thermocol box as shown in Fig. 1. This type of box is easily available in the market. Water is used as a coolant because it holds temperature for longer time than air can. An opening is created near the bottom of the box on one side which should be a little bigger in dimension



Fig. 6: Mounting of aluminium sheet over the thermocol box



Fig. 7: CPU heat-sink

than the thermoelectric module. An aluminium sheet is mounted over the opening from outside as shown in Fig. 6. Ensure that the sheet is firmly fixed and there is no leakage.

Use a silicone sealant or an adhesive to firmly fix the thermoelectric module over the aluminium sheet. The adhesive should not come between the sheet and the module. Now we need to mount a heat-sink over the thermoelectric module to dissipate heat from the hot side. Way of mounting the heat-sink will depend on the type of heat-sink you could obtain.

In our case, we used a spare CPU heat-sink which already had a nice base clip as shown in Fig. 7. The base clip was screwed and glued to the box. Then, the heat-sink was fitted on the base clip with its holding assembly. We also had the fan attached to the heat-sink. The refrigerator should be kept in a well-ventilated place enabling

the fan to blow good amount of air on the heat-sink.

Let the adhesive dry properly. For silicone, it takes about 40-50 minutes. Pour in a small amount of water inside the box and see if there is any leakage. In case of leakage, try putting more silicone to fill in the voids. Water level should be equal to the height of opening, so that maximum water comes into contact with the cold side of TEC module.

Water is a good conductor of heat only when it is in liquid form. Once it turns into ice, it will indeed block transfer of the heat. As said above, cold side can go well below 0°C and water near the opening will turn into ice in an hour or two. Once that happens, remaining water in the box will not get any colder. Hence we have to stop formation of ice near the opening.

Simple way of doing it is by using an aquatic air-pump. Pipe coming out from the air-pump is inserted in a box through a small hole at the top. The opening of the pipe from where the air blows out is facing the cold side of thermoelectric module as shown in Fig. 1. In this way, air bubbles do not let ice to form at the opening. Effectively, it creates a motion or water current and helps all of the water to reach the same temperature.

As there is water at the bottom, overall humidity inside the box goes high. Hence food items such as vegetables should be preferably stored in airtight containers or bags, as excessive moisture can ruin them. At least once a week, you should unplug the refrigerator and change water inside the box after cleaning it thoroughly.

Caution. Do not touch the hot side of the thermoelectric module in any case. ●

The author is an electronics engineer and hobbyist

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